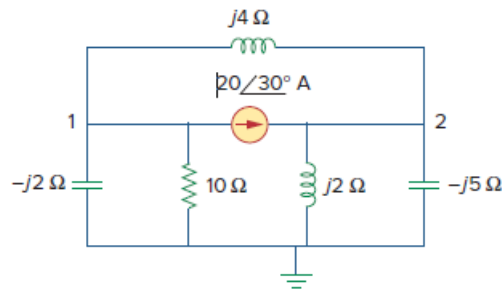


Problem

Calculate the voltage at nodes 1 and 2 in the circuit of Fig. 10.63 using nodal analysis.

Figure 10.63:



Step-by-step solution

Step 1 of 6

Refer to figure 10.63 given in the textbook.

Assume that the voltages of nodes 1 and 2 as V_1 and V_2 .

Apply Kirchhoff's current law at node 1.

$$\frac{V_1}{-j2} + \frac{V_1}{10} + 20\angle 30^\circ + \frac{V_1 - V_2}{j4} = 0$$

$$V_1 \left(\frac{-1}{j2} + \frac{1}{10} + \frac{1}{j4} \right) - V_2 \left(\frac{1}{j4} \right) = -20\angle 30^\circ$$

$$V_1 (0.1 + j0.25) - V_2 (-j0.25) = -17.3205 - j10$$

$$V_1 (0.1 + j0.25) + V_2 (j0.25) = -(17.3205 + j10) \dots\dots (1)$$

Step 2 of 6

Apply Kirchhoff's current law at node 2.

$$\frac{V_2}{-j5} + \frac{V_2}{j2} - 20\angle 30^\circ + \frac{V_2 - V_1}{j4} = 0$$

$$V_1 \left(\frac{-1}{j4} \right) + V_2 \left(\frac{-1}{j5} + \frac{1}{j2} + \frac{1}{j4} \right) = 20\angle 30^\circ$$

$$V_1 (j0.25) + V_2 (j0.2 - j0.5 - j0.25) = 20\angle 30^\circ$$

$$V_1 (j0.25) + V_2 (-j0.55) = (17.3205 + j10) \dots\dots (2)$$

Step 3 of 6

Recall equation (1).

$$V_1(0.1 + j0.25) + V_2(j0.25) = -(17.3205 + j10)$$

Recall equation (2).

$$V_1(j0.25) + V_2(-j0.55) = (17.3205 + j10)$$

The matrix form of equations (1) and (2) is,

$$\begin{bmatrix} 0.1 + j0.25 & j0.25 \\ j0.25 & -j0.55 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -17.3205 - j10 \\ 17.3205 + j10 \end{bmatrix}$$

Step 4 of 6

Calculate the determinants.

$$\begin{aligned} \Delta &= \begin{vmatrix} 0.1 + j0.25 & j0.25 \\ j0.25 & -j0.55 \end{vmatrix} \\ &= (0.1 + j0.25)(-j0.55) - (j0.25)^2 \\ &= 0.1375 - j0.055 - (-0.0625) \\ &= 0.1375 - j0.055 + 0.0625 \\ &= 0.2 - j0.055 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} -17.3205 - j10 & j0.25 \\ 17.3205 + j10 & -j0.55 \end{vmatrix} \\ &= (-17.3205 - j10)(-j0.55) - (17.3205 + j10)(j0.25) \\ &= -5.5 + j9.526275 - 17.3205 - j9.75 \\ &= -22.8205 - j0.223725 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 0.1 + j0.25 & -17.3205 - j10 \\ j0.25 & 17.3205 + j10 \end{vmatrix} \\ &= (0.1 + j0.25)(17.3205 + j10) - (-17.3205 - j10)(j0.25) \\ &= -0.76795 + j5.330125 - 2.5 + j4.330125 \\ &= -3.26795 + j9.66025 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

Delta 2 is alright, but Delta 1 is wrong i got -3 + 5.196i as delta 1

Step 5 of 6

Calculate node voltage V_1 .

$$V_1 = \frac{\Delta_1}{\Delta}$$

Substitute $(-22.8205 - j0.223725)$ for Δ_1 , $(0.2 - j0.055)$ for Δ .

$$\begin{aligned} V_1 &= \frac{-22.8205 - j0.223725}{0.2 - j0.055} \\ &= -105.794 - j30.212 \\ &= 110.023 \angle (-164.062^\circ) \text{ V} \end{aligned}$$

Therefore, the value of voltage V_1 is $110.023 \angle (-164.062^\circ) \text{ V}$.

Step 6 of 6

Calculate node voltage V_2 .

$$V_2 = \frac{\Delta_2}{\Delta}$$

Substitute $(-3.26795 + j9.66025)$ for Δ_2 , $(0.2 - j0.055)$ for Δ .

$$\begin{aligned} V_2 &= \frac{-3.26795 + j9.66025}{0.2 - j0.055} \\ &= -27.5399 + j40.7278 \\ &= 49.165 \angle (124.066^\circ) \text{ V} \end{aligned}$$

Therefore, the value of voltage V_2 is $49.165 \angle (124.066^\circ) \text{ V}$.